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LISTA 5 - MAPAS de Veitch-Karnaugh-ABCD

4. simplifique a expressão booleana pelo mapa de Karnaugh:

$$F(A,B,C) = \overline{A}\,\overline{B}\,\overline{C} + \overline{A}\,\overline{B}C + \overline{A}B\overline{C} + AB\overline{C}$$

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

$\overline{A}\,\overline{B}\,\cancel{C}$ 0 0 $\cancel{0}$ $\overline{A}\,\overline{B}\,\cancel{C}$ 0 0 $\cancel{1}$

A \ BC	00	01	11	10
0	1	1	0	1
1	0	0	0	1

$\cancel{A}B\overline{C}$ $\cancel{0}$ 1 0 → $\cancel{A}B\overline{C}$ $\cancel{1}$ 1 0

R: $\boxed{\overline{A}\,\overline{B} + B\overline{C}}$

6. Obtenha a expressão booleana simplificada pelo mapa de Karnaugh.

A	B	C	D	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

AB \ CD	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
00	1	0	0	1
01	1	1	1	1
11	1	0	1	1
10	1	0	0	1

$\bar{D} + \bar{A}B + CD$

$\bar{A}$ 11 B 11

R: $\bar{D} + \bar{A}B + CD$

7. Obtenha a expressão booleana simplificada pelo mapa de Karnaugh. $F = \bar{A}\bar{B}C + \bar{A}BC + ABC + A\bar{B}C$

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$ 0	0	1	1	0
$\bar{A}$ 1	0	1	1	0

R: C.,,

$F = C$

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	0	1	1	0
A 1	0	1	1	0

$$\boxed{F = C}$$